espark-aman Update README.md 3e1aa87 · 1 minute ago

120 lines (61 loc) · 3.78 KB

ESP32-C3 Deep Sleep Implementation

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Project Requirements

Implement a wake-on-movement functionality to an existing esp32-based open source project, the open source project that needs to be expanded is shared under docs. Device should sleep after 5 minutes of little-to-no-movement scenario, and wake upon movement. Upon wake up, the charge indicator (red LED on XIAO esp32c3) should flash three times. Components are:

- Xiao ESP32-C3 for MCU
- BMI270/LSM6DSV16X for IMU.

Step 1 : Install the ESP-IDF(v5.2.1) Extension for VS Code

1- Open Visual Studio Code (VS Code):

Launch VS Code on your computer.

2- Go to the Extensions View:

Press Ctrl+Shift+X (Cmd+Shift+X on macOS) to open the Extensions view, or click on the Extensions icon located in the left sidebar of VS Code.

3- Search for ESP-IDF:

In the Extensions search bar, type ESP-IDF.

4- Install the ESP-IDF Extension:

Find ESP-IDF by Espressif Systems and click Install.

Step 2 : Install ESP-IDF Tools.

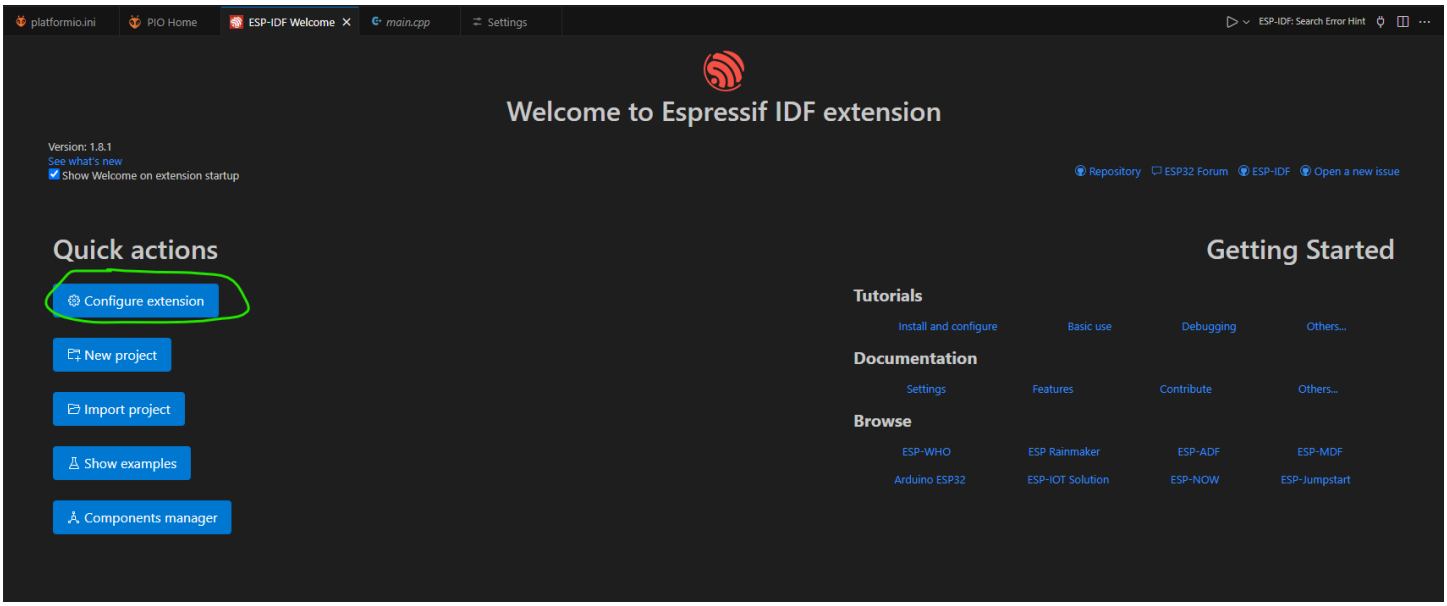
1- Open the Command Palette:

Press Ctrl+Shift+P (Cmd+Shift+P on macOS) to open the Command Palette.

2- Run ESP-IDF: Configure ESP-IDF:

1- Type ESP-IDF: Configure ESP-IDF extension and select it from the list.

2- This command will guide you through the process of downloading and installing the required tools.



3- Select the Installation Path:

When prompted, select a directory where the ESP-IDF tools will be installed (e.g., C:\esp-idf or a directory of your choice).

4- Install Dependencies:

The extension will ask to install Python dependencies and other necessary tools. Follow the prompts to install Python packages, Git, CMake, Ninja, and other dependencies required for ESP-IDF development

5- Verify the Installation:

Once the installation is complete, you can verify that everything is set up correctly by opening a new terminal in VS Code and typing

```
idf.py --version
```

Step 3 : Installation Of PlatformIO IDE(v3.3.3).

Install PlatformIO IDE from the extension of VS Code. PlatformIO IDE will automatically install in VS Code and may prompt you to reload or restart vs-code after installation.

1- Launch PlatformIO IDE

Once the installation is complete, you can open PlatformIO IDE from within Visual Studio Code:

2- Open PlatformIO Home:

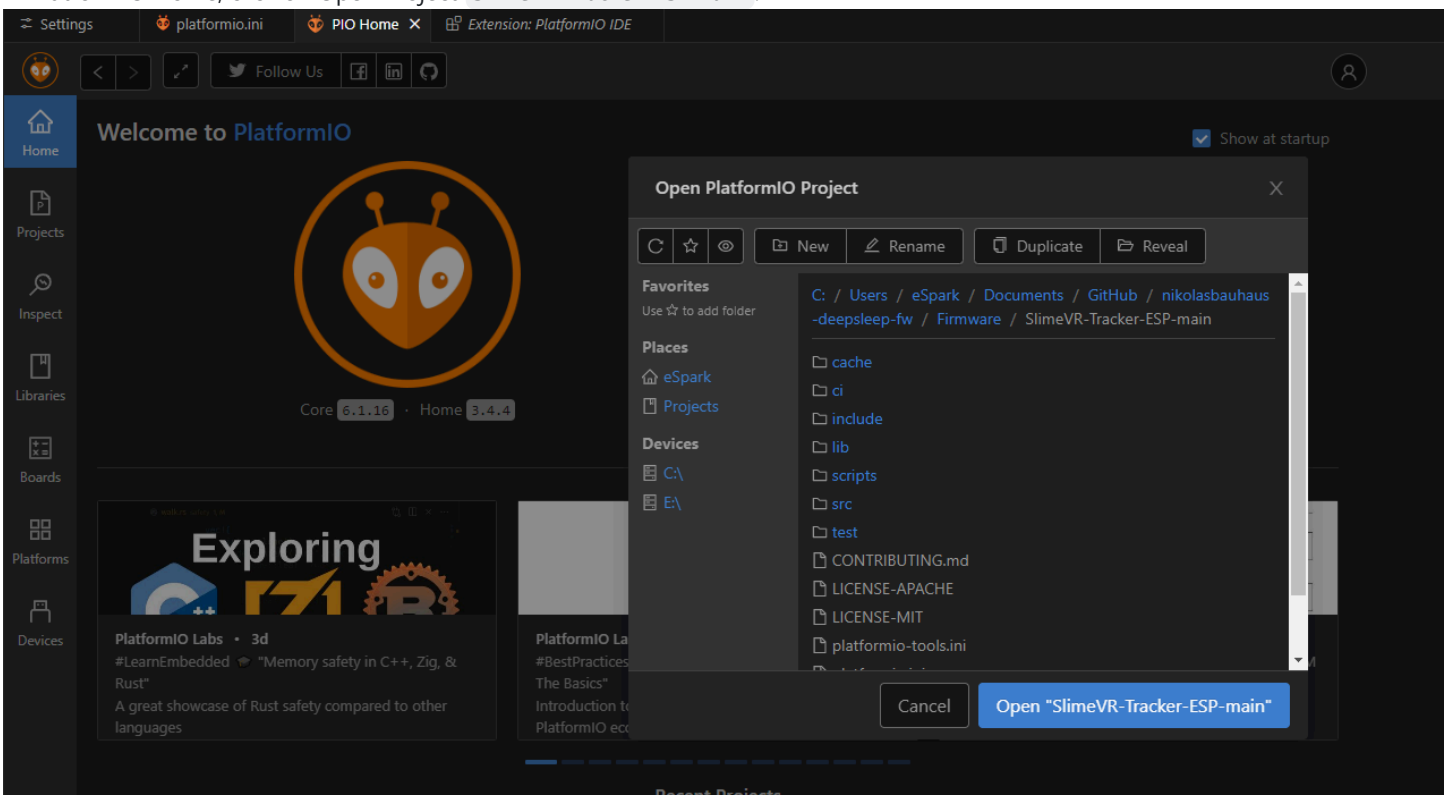
Press Ctrl+Shift+P (or Cmd+Shift+P on macOS) to open the Command Palette. Type PlatformIO: Home and select it. Alternatively, you can also open the PlatformIO Home by clicking the PlatformIO icon (alien head) in the left sidebar of VS Code.

Set Up Your PlatformIO Project

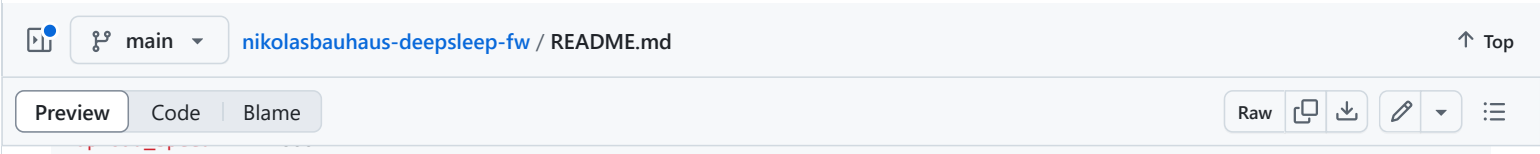
Once PlatformIO is installed and running, you can set up your project:

1- Open the Project:

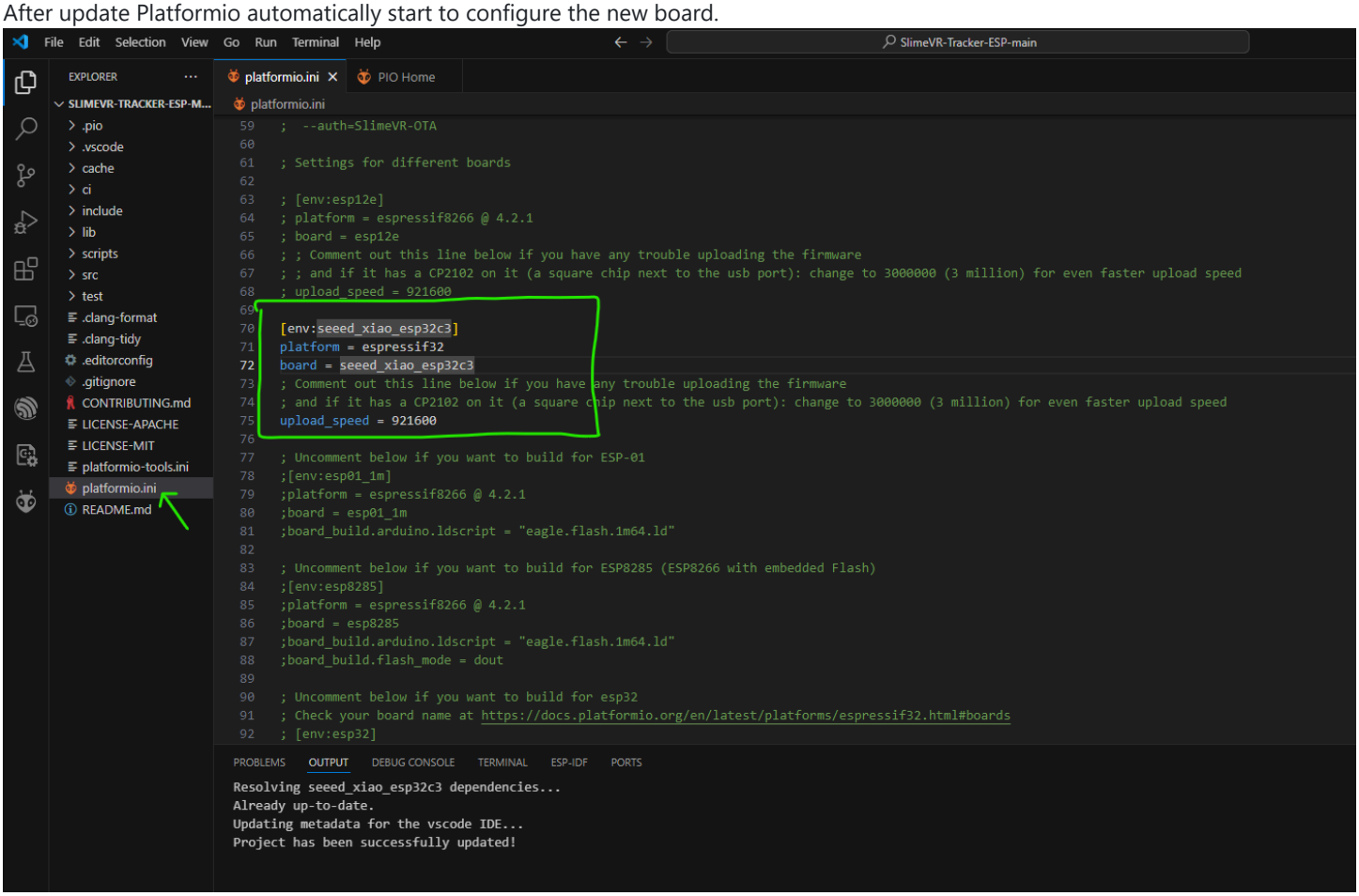
In PlatformIO Home, click on Open Project `SlimeVR-Tracker-ESP-main`.



2- Update the board in `platformio.ini` file.



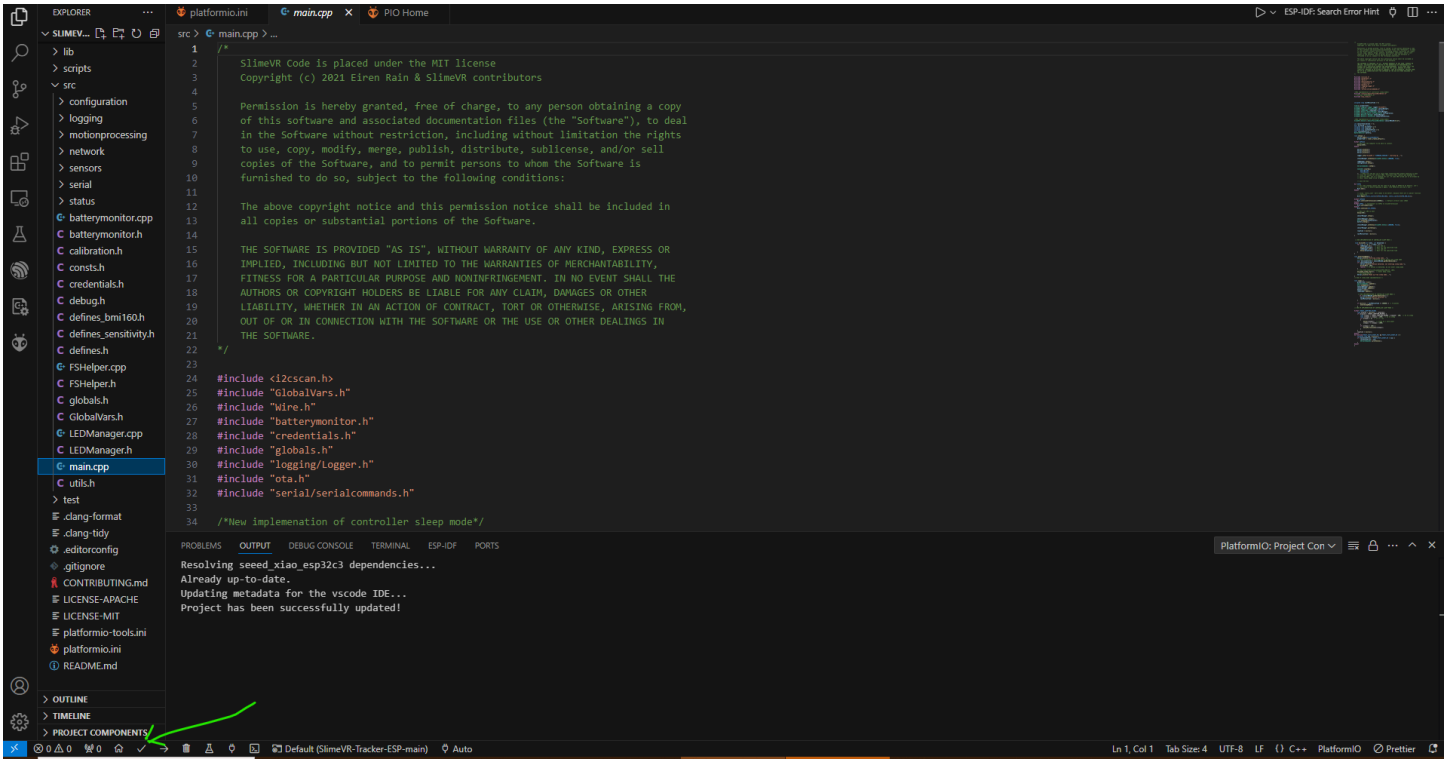
After update Platformio automatically start to configure the new board.



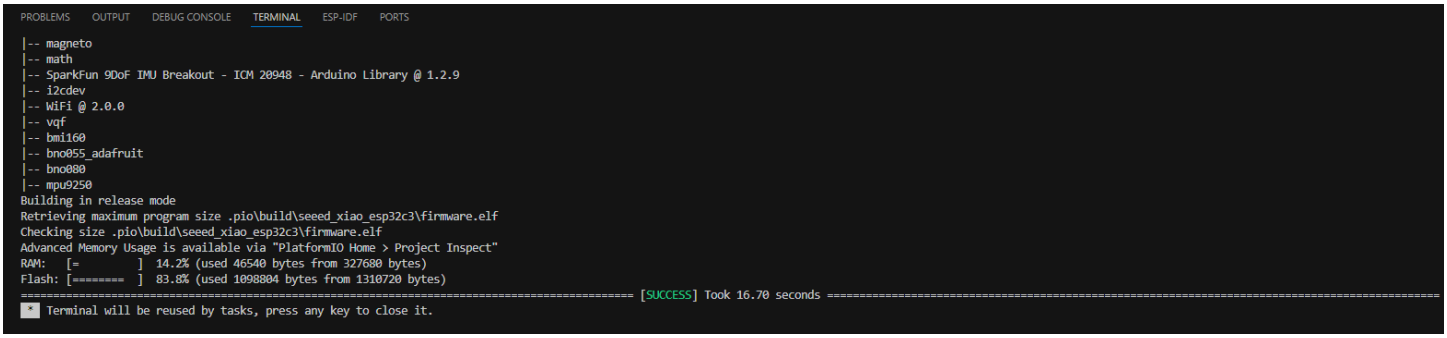
Step 4 : Compilation Process.

1- Navigate to the main.cpp File.

After successfully adding your code, open the main.cpp file in your project. Click on tick to compile, it will take some time for compilation.

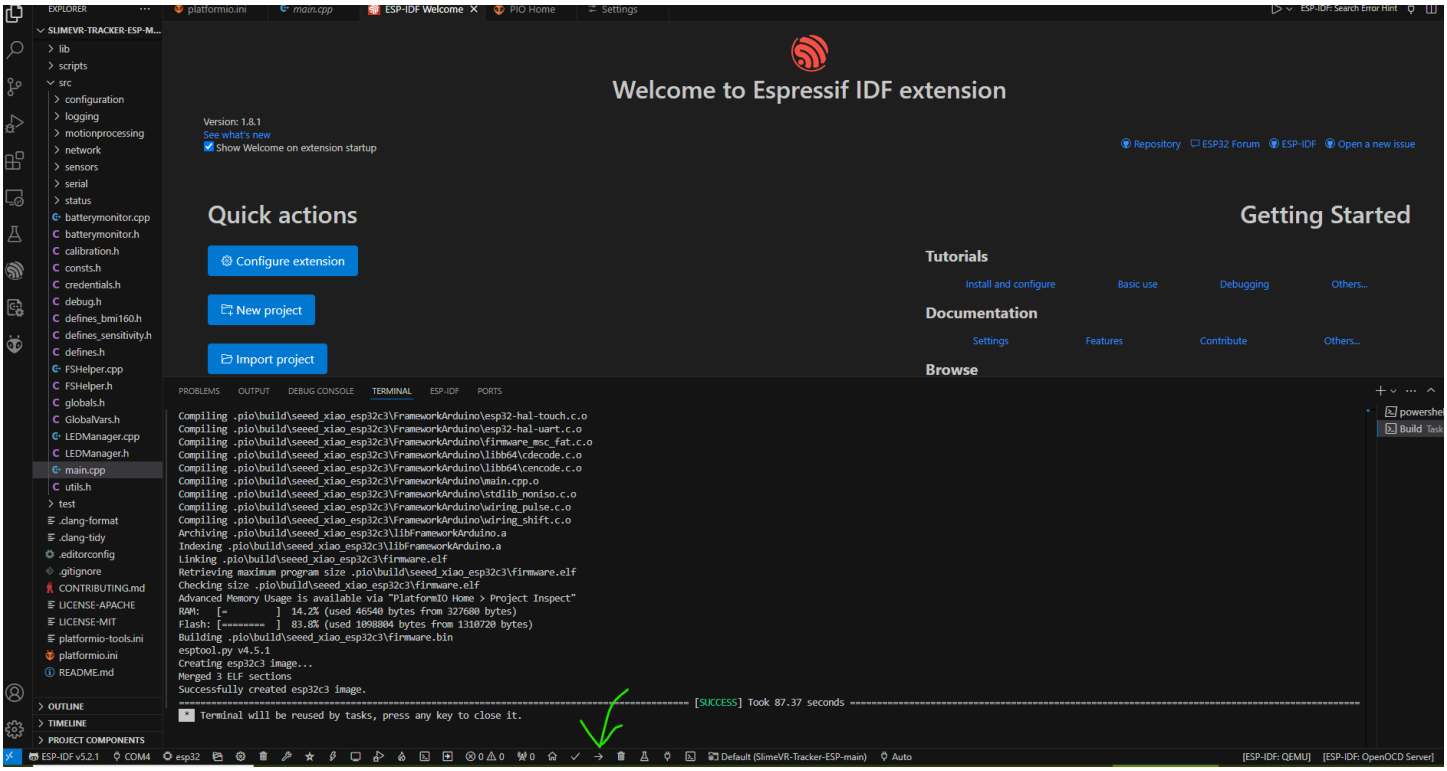


2- Successfully Compile.



Step 5 : Flashing.

Click on arrow to flash.


Welcome to Espressif IDF extension
Quick actions
Configure extension
New project
Import project
Getting Started
Tutorials
Documentation
Browse
Compiling .pio\build\seeed_xiao_esp32c3\FrameworkArduino\esp32-hal-touch.c.o
Compiling .pio\build\seeed_xiao_esp32c3\FrameworkArduino\esp32-hal-uart.c.o
Compiling .pio\build\seeed_xiao_esp32c3\FrameworkArduino\firmware_msc_fat.c.o
Compiling .pio\build\seeed_xiao_esp32c3\FrameworkArduino\libb64\cdecode.c.o
Compiling .pio\build\seeed_xiao_esp32c3\FrameworkArduino\libb64\cencode.c.o
Compiling .pio\build\seeed_xiao_esp32c3\FrameworkArduino\main.cpp.o
Compiling .pio\build\seeed_xiao_esp32c3\FrameworkArduino\stdlib_noniso.c.o
Compiling .pio\build\seeed_xiao_esp32c3\FrameworkArduino\wiring_pulse.c.o
Compiling .pio\build\seeed_xiao_esp32c3\FrameworkArduino\wiring_shift.c.o
Archiving .pio\build\seeed_xiao_esp32c3\libFrameworkArduino.a
Indexing .pio\build\seeed_xiao_esp32c3\libFrameworkArduino.a
Linking .pio\build\seeed_xiao_esp32c3\firmware.elf
Retrieving maximum program size .pio\build\seeed_xiao_esp32c3\firmware.elf
Checking size .pio\build\seeed_xiao_esp32c3\firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM: [=] 14.2% (used 46540 bytes from 327680 bytes)
Flash: [========] 83.8% (used 1098804 bytes from 1310720 bytes)
Building .pio\build\seeed_xiao_esp32c3\firmware.bin
esptool.py v4.5.1
Creating esp32c3 image...
Merged 3 ELF sections
Successfully created esp32c3 image.
[SUCCESS] Took 87.37 seconds
Terminal will be reused by tasks, press any key to close it.